

GROUP WORK PROJECT # 2
MANAGEMENT
GROUP NUMBER: 14996

MScFE 652: PORTFOLIO

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MScFE 652 Portfolio Management

Group Work Project #2

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Abstract

This project is the result of development of project MScFE 652 Portfolio Management.

AI Usage Disclosure

Artificial intelligence tools were used for:

- Verifying the accuracy of data sourcing and references;
- Validating the mathematical implementation of used algorithms
- Assisting in code optimization and debugging to ensure reproducibility of results.

All interpretations, conclusions, and original content presented in this report were developed independently by the authors, and all AI-assisted work has been reviewed and verified for accuracy.

Table 1: Project snapshot

Item	Value
Asset universe	12 assets across equities, gold, treasuries, and REITs (Table 3).
Training window	Jan 1, 2024 – Sep 30, 2025 (438 trading days).
Test window	Oct 1, 2025 – Dec 31, 2025 (63 trading days).
Main constraint	Fully invested; no shorts; $0 \leq w_i \leq 0.20$.
Sizing choices	Kelly (100%), Double-Kelly (200%), Half-Kelly (50%), plus Denoised-Kelly, Detoned-Kelly, Denoised+Detoned-Kelly, and HRP in the final ML comparison.
Main test result	Double-Kelly returns the most, but HRP has the strongest Sharpe ratio and the smallest drawdown in the final seven-strategy comparison (Table 30).
Next improvement idea	Cross-validate the Kelly family and compare denoising, clustering, and detoning in one integrated workflow (Steps 3–6).

1 Introduction

This project is about building a simple growth-focused portfolio and then checking how it holds up out-of-sample. Table 2 gives the main ideas without a long setup.

Table 2: Introduction summary (plain language)

Topic	What it means here	Reference
Portfolio choice	Pick weights to balance return and risk in a disciplined way.	Markowitz (1952)
Kelly idea	Focus on long-run growth by working with log returns.	Kelly (1956)
Practical constraints	Keep weights realistic: fully invested, no shorts, 20% cap per asset.	Step 1
Why denoising	Covariance estimates can be noisy, and that noise can push weights around.	Laloux et al. (1999); López de Prado (2020)
What we report	Build weights in-sample, then compare Kelly/Double/Half out-of-sample.	Steps 1–2

2 Asset Universe and Data

2.1 Portfolio Construction

Table 3: Portfolio Asset Universe

Asset	Name	Sector / Asset Class
AAPL	Apple Inc.	Technology
MSFT	Microsoft Corp.	Technology
AMZN	Amazon.com Inc.	Consumer Discretionary
JPM	JPMorgan Chase & Co.	Financials
JNJ	Johnson & Johnson	Healthcare
XOM	Exxon Mobil Corp.	Energy
PG	Procter & Gamble Co.	Consumer Staples
NEE	NextEra Energy Inc.	Utilities
BRK-B	Berkshire Hathaway B	Diversified Conglomerate
GLD	SPDR Gold Shares ETF	Commodities
TLT	iShares 20+ Year Treasury ETF	Fixed Income
VNQ	Vanguard Real Estate ETF	Real Estate

Table 4: Data source and return construction

Data item	How it was handled
Prices	Daily adjusted close, downloaded from Yahoo Finance.
Returns	Log-returns $r_{i,t} = \ln(P_{i,t}/P_{i,t-1})$ computed for each asset and day.

2.2 Data Periods

Table 5: Sample windows and setup

Piece	Dates	Size
In-sample (training)	Jan 1, 2024 – Sep 30, 2025	438 trading days
Out-of-sample (testing)	Oct 1, 2025 – Dec 31, 2025	63 trading days
Risk-free rate	4.5% p.a. (annualised)	Used for excess return

2.3 Asset Correlations

Figure 1 shows the correlation matrix over the training period that are summarised in Table 7, and the broader cross-asset dynamics support a multi-asset allocation strategy:

Table 6: What correlation say (training period)

Pattern	What it suggests
Tech moves together	Tech names show stronger correlation, so they can behave like one single risk bucket.
GLD/TLT hedge role	GLD and TLT are uncorrelated (or slightly negative) with equities, so they can help diversification.
Room for diversification	Correlations are not all high, so mixing assets can still reduce portfolio swings.

Table 7: Key Pairwise Correlations (Training Period: January 1, 2024 – September 30, 2025)

Pair	Comment	Correlation
AAPL-MSFT	Technology sector peers	0.50
AMZN-MSFT	Technology/consumer discretionary sector peers	0.67
GLD-VNQ	Gold commodity vs. broad equity market	$\approx 0 / < 0$
TLT-VNQ	Long-duration Treasuries vs. growth sectors	$\approx 0 / < 0$

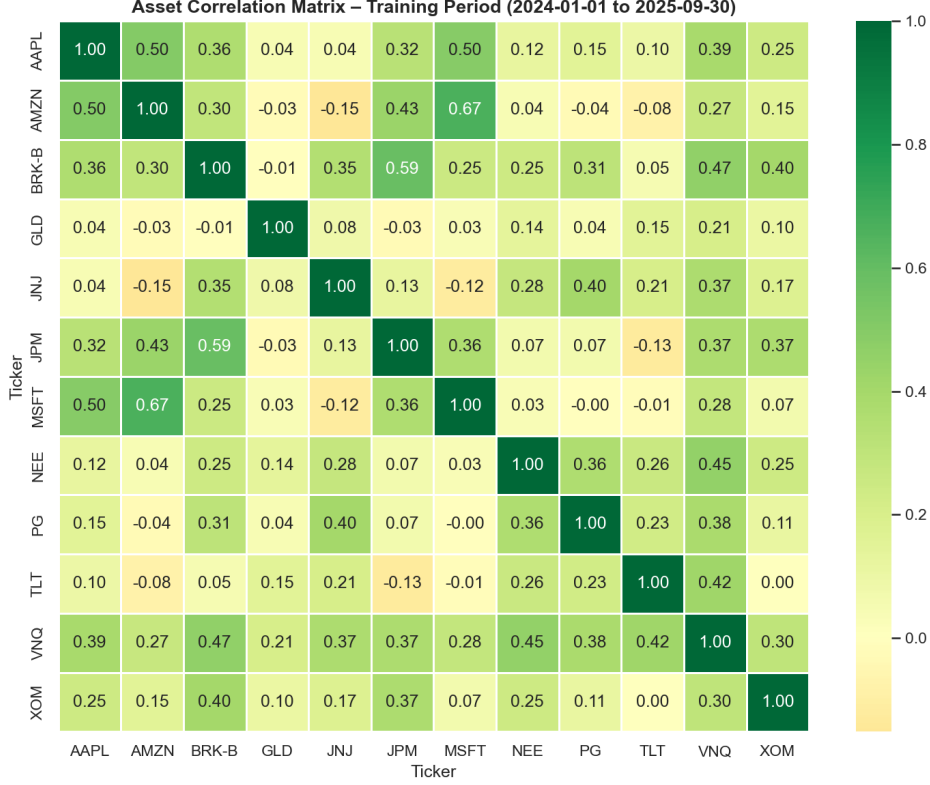


Figure 1: Asset correlation matrix over the training period (January 1, 2024 – September 30, 2025). Darker green indicates higher positive correlation.

3 Step 1: Optimal Kelly Portfolio

3.1 Theoretical Framework

We used a standard Kelly-style mean–variance objective on daily log-returns (Kelly 919–920):

$$\max_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\mu} - \frac{1}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$$

Table 8 summarises our setup and the constraints we used.

Table 8: Kelly optimisation setup (what each piece means)

Piece	Meaning
$\mathbf{w}$	Portfolio weights across the N assets.
$\boldsymbol{\mu}$	Daily excess mean log-returns (net of the risk-free rate), estimated in-sample.
$\boldsymbol{\Sigma}$	Daily covariance matrix of log-returns, estimated in-sample.
Unconstrained solution	$\mathbf{w}^* = \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$ (Thorp 389).
Constraints used here	Fully invested; no shorting; per-asset cap of 20% (shown below).
Numerical solve	Quadratic program solved with <code>cvxpy</code> (CLARABEL).

The unconstrained analytical solution is (Thorp 389):

$$\mathbf{w}^* = \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$$

We add practical constraints to keep the weights realistic:

$$\sum_{i=1}^N w_i = 1 \quad (\text{fully invested})$$

$$0 \leq w_i \leq 0.20 \quad \forall i \quad (\text{no short-selling; 20\% cap per asset})$$

We used `cvxpy` convex optimisation library with the CLARABEL solver to solve numerically this quadratic problem.

3.2 Optimal Weights

The constrained Kelly optimisation produces the weights shown in Table 9 and Figure 2.

Table 9: Constrained Kelly Portfolio Weights ($\leq 20\%$ per asset)

Asset	Kelly Weight (%)	Double-Kelly Weight (%)
AMZN	20.00	40.00
BRK-B	20.00	40.00
GLD	20.00	40.00
JPM	20.00	40.00
MSFT	20.00	40.00
AAPL	0.00	0.00
JNJ	0.00	0.00
NEE	0.00	0.00
PG	0.00	0.00
TLT	0.00	0.00
VNQ	0.00	0.00
XOM	0.00	0.00
Total	100.00	200.00

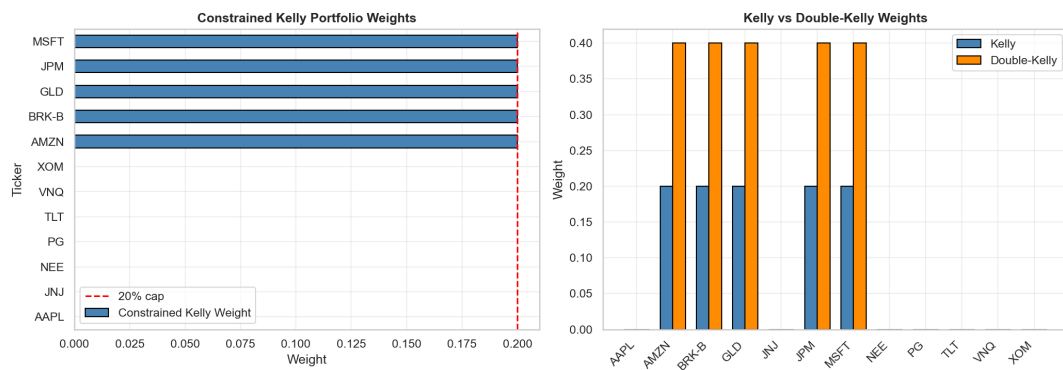


Figure 2: Left: Constrained Kelly weights with the 20% cap (dashed red line). Right: Side-by-side comparison of Kelly and Double-Kelly allocations.

3.3 Interpretation of Weights

The solution ends up being very “all or nothing”: a few names at the 20% cap and the rest at zero. Table 10 explains why that happens, and Table 11 gives the simple rationale by asset group.

Table 10: Why the weights look “bang-bang” in this case

What we see	Simple reason
Five assets at 20%	The unconstrained Kelly solution wants to concentrate on the strongest contributors, so once the cap binds it fills up the top picks first.
Many assets at 0%	Some assets do not improve the risk/return trade-off enough after correlations are considered, so the optimiser leaves them out under the cap.
Clustering at the bounds	With a hard cap and no shorting, the optimiser often ends up pushing weights to the extremes (0% or 20%) instead of spreading them evenly.

Table 11: Rationale for Portfolio Weight Allocations

Group	Tickers	Explanation
Included (20% weight)	AMZN, MSFT	Did well over the period, especially compared with the amount of day-to-day swings
	JPM	Benefited from the higher interest-rate backdrop and held up better than most
	BRK-B	Generally steadier than the rest and still delivered solid returns
	GLD	Helped balance the equity risk since it did not move in step with the stocks
Excluded (0% weight)	JNJ, NEE, PG, XOM, VNQ	Nothing wrong with them, but they did not add enough return or diversification in this setup
	TLT	Did not add much extra protection on top of GLD, and long bonds were hurt as rates moved up in 2024–2025

4 Step 2: Out-of-Sample (test dataset) Performance Comparison

4.1 Strategy and Performance Metrics Overview

Tables 12 and 13 define the three sizing choices and the metrics used for comparison.

Table 12: Out-of-Sample Strategy Definitions

Strategy	Gross Notional Exposure	Description
Kelly	100%	Just the basic constrained allocation using the raw weights from Table 9, all money put in with no borrowing to add extra positions.
Double-Kelly	200%	Levered version where every Kelly weight is doubled, so the portfolio is 2:1 on gross exposure.
Half-Kelly	50%	More cautious version where every Kelly weight is cut in half and the rest is held in the risk-free asset.

Table 13: Performance Metric Definitions

Metric	Formula	Interpretation
Cumulative Return	$R_{cum} = e^{\sum_t r_t} - 1$	How much the whole portfolio grew (or shrank) in percentage terms over the whole test window, figured by adding up all the daily log returns we tracked.
Sharpe Ratio	$S = \frac{\bar{r}_{excess}}{\sigma_r} \sqrt{252}$	Annual number that shows if returns were worth the risk taken, compares average extra returns above the 4.5% risk-free rate to how much daily returns bounced around, then scaled to a full year.
Maximum Drawdown (MDD)	$MDD = \min_t \frac{V_t - M_t}{M_t}, \quad M_t = \max_{s \leq t} V_s$	Biggest single drop from a high point to a low point the portfolio saw during the test, basically the worst loss anyone holding this would've had to sit through at any point.
95%-CVaR (daily)	Empirical mean of the worst 5% of daily returns	Average loss on the really bad trading days, the 5% worst ones, gives a sense of how bad those terrible days usually were, not just the single worst one.

4.2 Results

Table 14: Out-of-sample performance – October 1 – December 31, 2025

Strategy	Cumulative Return (%)	Sharpe Ratio	Max Drawdown (%)	95%-CVaR daily (%)
Kelly	2.39	0.40	−5.54	−1.44
Double-Kelly	4.83	0.59	−10.77	−2.88
Half-Kelly	1.19	0.02	−2.81	−0.72

Note: CVaR and MDD are negative values (losses). Risk-free rate: 4.5% p.a.

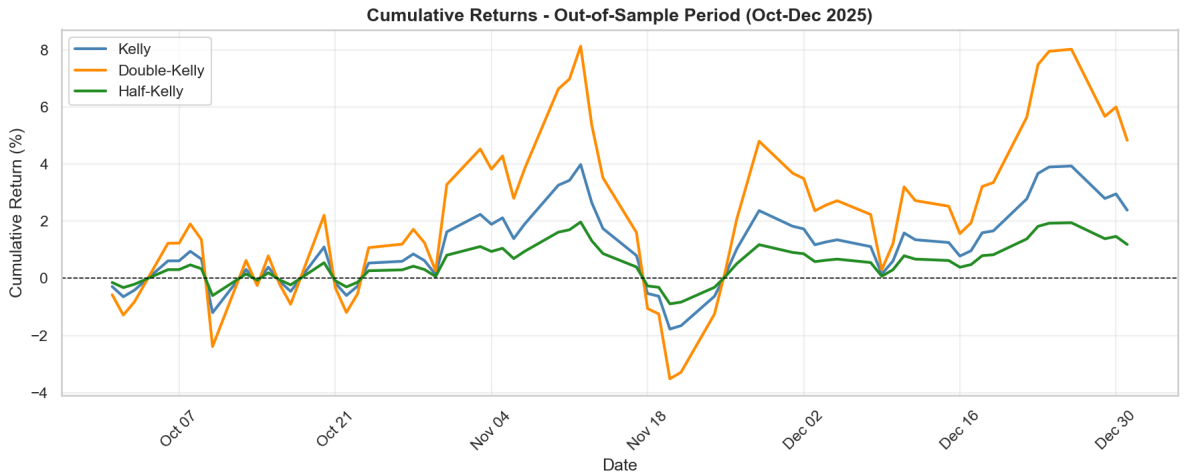


Figure 3: Cumulative returns of the three strategies – out-of-sample period (October 1 – December 31, 2025).

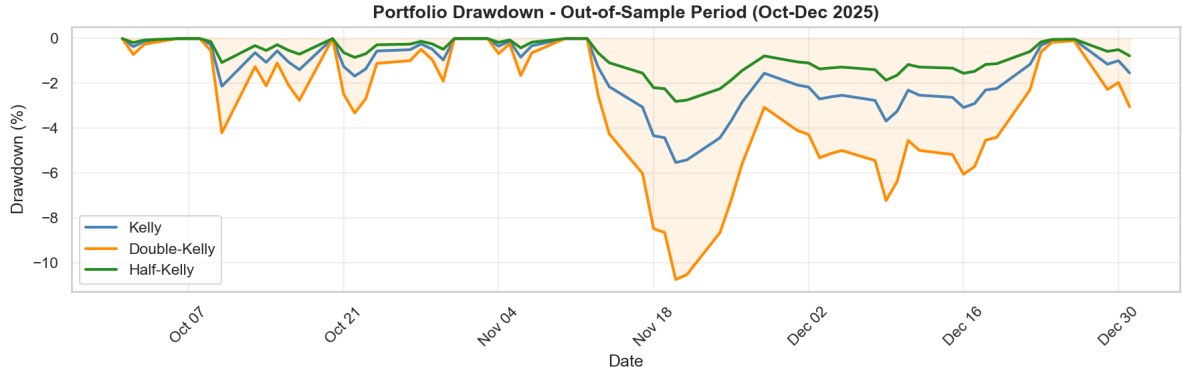


Figure 4: Portfolio drawdown paths – out-of-sample period. The shaded area highlights the Double-Kelly drawdown, which peaked at -10.77% around mid-November 2025.

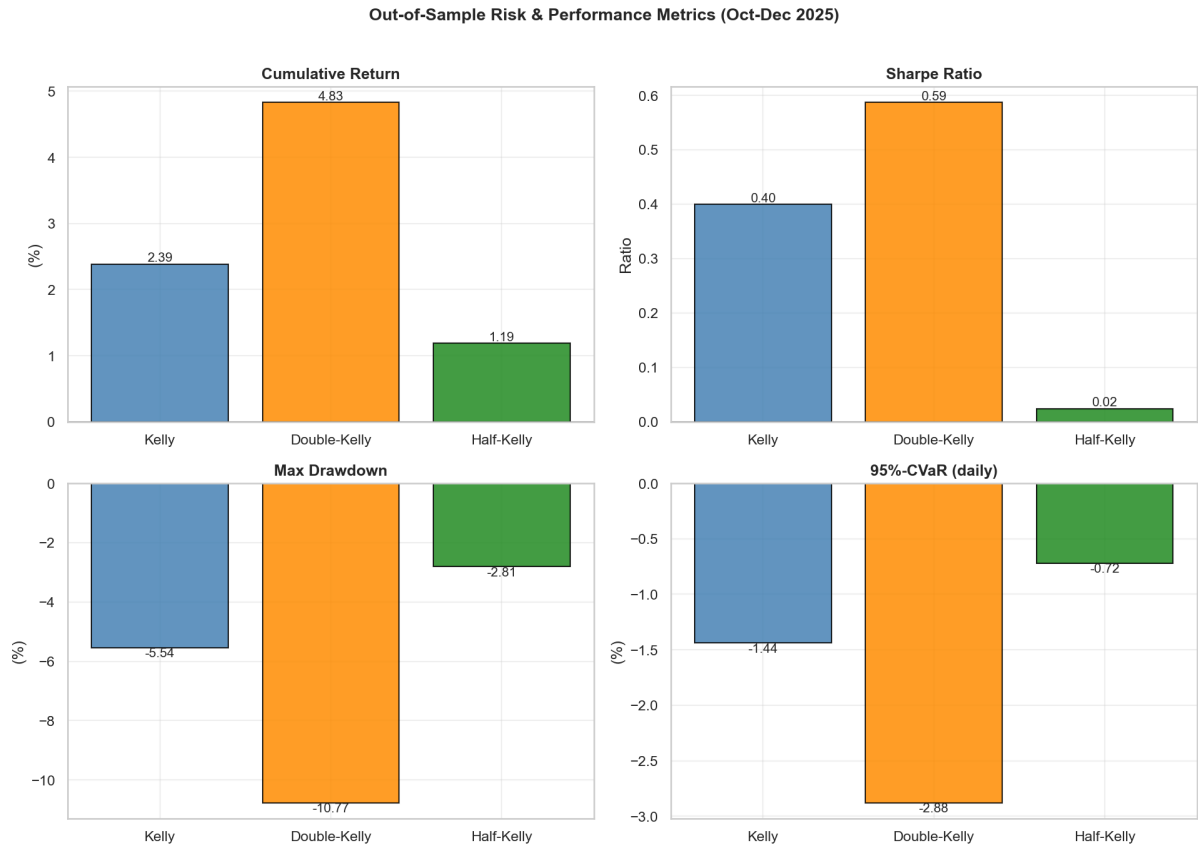


Figure 5: Summary of the four risk and performance metrics for each strategy.

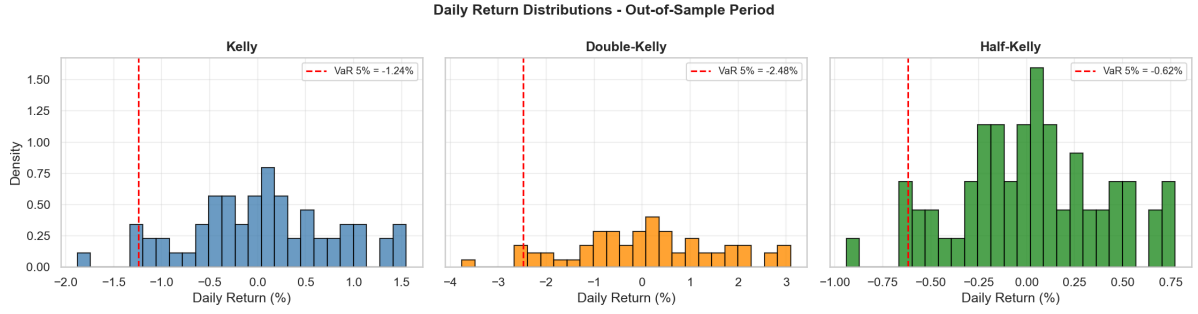


Figure 6: Empirical daily return distributions and 5th percentile (VaR) thresholds for each strategy – out-of-sample period.

4.3 Analysis and Discussion

The results in Table 14 are easier to read when we translate the numbers into a few simple takeaways. Table 15 summarises the main points, and Table 16 gives a practical interpretation.

Table 15: Out-of-sample results: quick interpretation

Metric	Kelly	Double-Kelly	Half-Kelly	Explanation
Cumulative return	2.39%	4.83%	1.19%	Double-Kelly is roughly “twice the exposure, roughly twice the result” over this short window (Thorp 395–396).
Sharpe ratio	0.40	0.59	0.02	In practice, the risk-free adjustment matters: when returns are small, subtracting the risk-free rate can noticeably change the ratio, especially over a short test period.
Max drawdown	−5.54%	−10.77%	−2.81%	Losses scale up with leverage too. The worst dip for Double-Kelly is about twice Kelly, and it happened around mid-November 2025 during a broad equity sell-off.
95%-CVaR (daily)	−1.44%	−2.88%	−0.72%	On the worst 5% of days, Double-Kelly loses about twice as much as Kelly, which is the key cost of leverage.

Table 16: Practical takeaways from the three leverage choices

Investor style	Suggested strategy	Brief reason
Balanced risk appetite	Kelly	Keeps decent upside (2.39%) while drawdowns stay more manageable than the leveraged version.
High risk tolerance and leverage capacity	Double-Kelly	Higher return, but you must be comfortable with larger drawdowns and worse tail losses.
More cautious sizing / uncertainty is high	Half-Kelly	Smaller drawdowns, but risk-adjusted performance can look weak over short windows once the risk-free rate is accounted for (López de Prado 42–44).

5 Step 3: K-fold Cross-Validation Backtesting

5.1 Walk-forward K-fold design

Instead of judging the portfolio rules on one out-of-sample window only, we rotate the test window across the full 2024–2025 sample and re-estimate the model each time. The standard random K-fold procedure is not appropriate for financial data because shuffling would let future observations leak into the training set. We used a walk-forward variant in which the training window always ends before the test window begins, and we add a 5-day embargo at the boundary to reduce contamination from short-run autocorrelation.

Table 17: Step 3 backtesting design

Item	Implementation in the script
Dataset used	Full sample from Jan 3, 2024 to Dec 31, 2025 (501 daily log-return observations after differencing).
Number of folds	5 walk-forward folds.
Initial training size	86 observations.
Test size per fold	83 observations.
Embargo	5 trading days between training end and test start.
Strategies compared	Kelly, Double-Kelly, and Half-Kelly.
Metrics reported	Cumulative return, Sharpe ratio, Sortino ratio, Maximum Drawdown, and 95%-CVaR.

5.2 Cross-validation results

Double-Kelly has the highest mean return and the highest mean Sharpe ratio across folds, but it is also the least stable. Its standard deviation across folds is almost as large as its mean Sharpe ratio, and its average drawdown and tail loss are materially worse than the alternatives. Kelly sits in the middle, while Half-Kelly gives the most conservative profile, with smaller losses but also weaker risk-adjusted payoff.

Table 18: Step 3 cross-validation summary: mean (standard deviation) across 5 folds

Strategy	Cum. Return (%)	Sharpe	Sortino	Max Drawdown (%)	95%-CVaR (%)
Kelly	4.42 (3.19)	0.83 (0.95)	0.78 (0.91)	−7.44 (4.73)	−1.92 (1.01)
Double-Kelly	9.12 (6.68)	1.01 (0.99)	0.95 (0.95)	−14.15 (8.50)	−3.84 (2.02)
Half-Kelly	2.18 (1.56)	0.46 (0.87)	0.43 (0.81)	−3.82 (2.49)	−0.96 (0.50)

Note: values are mean (standard deviation) across the 5 walk-forward folds.

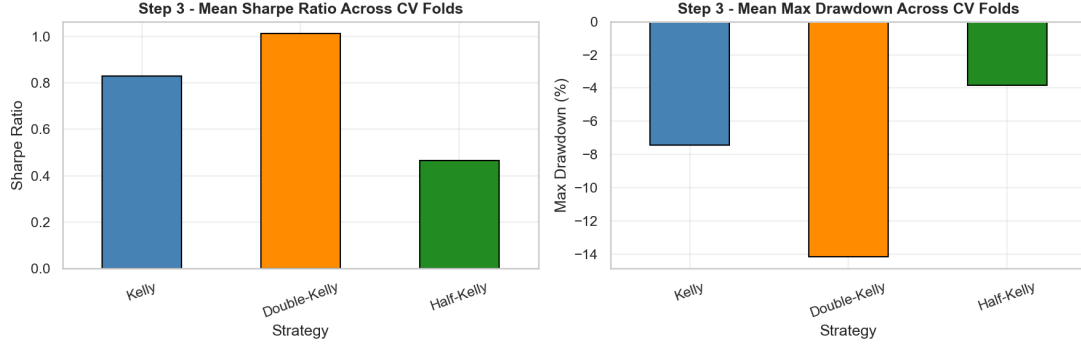


Figure 7: Step 3 cross-validation summary. Left: mean Sharpe ratio across folds. Right: mean maximum drawdown across folds.

5.3 Interpretation

This pattern is close to the logic in previous steps. Leverage can help when the estimated edge is right, so Double-Kelly often posts the highest fold return, but the same leverage also magnifies estimation error and tail losses. In practical terms, the walk-forward backtest says that Double-Kelly is the most aggressive choice, Kelly is the middle-ground choice, and Half-Kelly is the most robust if parameter uncertainty is the main concern.

6 Step 4: Improvements Using Denoising, Clustering, and Deton- ing

6.1 The Estimation Error Problem in Covariance Matrices

Here we estimated the covariance matrix Σ , and found that it is noisy. Estimates are noisy, and that noise can move weights around. Table 19 puts the idea in the context of our data.

Table 19: Estimation noise: what it means for this project

Piece	Plain meaning
What is being estimated	Expected returns μ and covariance Σ from historical data.
Why it is a problem	Random variation can look like a pattern, and the optimiser may chase it.
Why covariances matter a lot	The Kelly-style solution depends heavily on Σ^{-1} , which can amplify estimation noise.
Why this study still makes sense	With $N = 12$ and $T = 438$, estimates are workable but still noisy enough that cleaning Σ can help (Laloux et al. 1467).

6.2 Random Matrix Theory and the Marcenko-Pastur Distribution

RMT gives a rule of thumb for which eigenvalues look like noise, based on $q = N/T$ (López de Prado 58; Laloux et al. 1467–1468).

$$\lambda^{\pm} = \sigma^2 \left(1 \pm \frac{1}{\sqrt{q}} \right)^2$$

Table 20: RMT quantities used in this study

Quantity	How we use it
$q = N/T$	Ratio that controls the noise band for eigenvalues. Here $q = 12/438 \approx 0.0274$.
λ^+	Upper bound of the “mostly noise” region. Here $\lambda^+ \approx 1.37$.
Interpretation	Eigenvalues at/below λ^+ are treated as noise-like; bigger eigenvalues are treated as structure worth keeping.

6.3 The Denoising Procedure

Table 21 summarises the standard Marcenko-Pastur denoising recipe (López de Prado 60–63).

Table 21: Denoising procedure (Marcenko-Pastur): plain summary

Step	What you do
1	Build the sample correlation matrix $\hat{\mathbf{C}}$ from log-returns.
2	Compute eigenvalues and eigenvectors: $\hat{\mathbf{C}} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$.
3	Use λ^+ to decide which eigenvalues look like noise (those at or below λ^+).
4	Replace the “noise” eigenvalues with a single average level (chosen so the total variance is preserved), then rebuild a denoised $\tilde{\mathbf{C}}$.
5	Convert back to a covariance matrix using the original asset standard deviations.

6.4 Benefits for Kelly Optimisation

Applying denoising before solving the constrained Kelly problem is mainly about making the inputs less fragile. Table 22 summarises the practical benefits.

Table 22: Why denoising can help the Kelly optimisation

Benefit	Plain interpretation
Less overfitting in $\mathbf{\Sigma}^{-1}$	The optimiser can overreact to small, noisy eigenvalues because inversion amplifies them. Denoising reduces this effect and usually leads to more stable weights (López de Prado 64–65).
Better numerical stability	A better-conditioned covariance matrix makes optimisation and inversion less sensitive to small data changes.
Less “whipsaw” turnover	Weights tend to move less from one rebalance to the next, which matters once you consider transaction costs.
Empirical support	Denoised covariances often improve out-of-sample Sharpe ratios in practice, especially when T/N is not very large (López de Prado 66).

6.5 Application to Our Portfolio

If we denoise the covariance estimates for this 12-asset universe, the main expectation is that the allocation would be less “all-in” on the same five names at the 20% cap. In simple terms, denoising can make the optimiser less confident in very concentrated bets. Table 23 states the likely direction of change.

Table 23: Expected effect of denoising in this study (directional)

What would change	What we expect to see
Weight concentration	A less extreme “five-at-the-cap” pattern, with some weight spreading across more assets.
Previously excluded assets (e.g., TLT, VNQ)	They may look more helpful once noisy covariance effects are reduced, so they could receive small positive weights in the constrained solution.
Tail risk measures (MDD, CVaR)	Likely lower, since the portfolio may be less concentrated in correlated equity exposures.
Raw cumulative return	Possibly a bit lower, since reducing concentration can give up some upside when the concentrated names keep performing.

Steps 5 and 6 apply this and compare the denoised version out-of-sample.

6.6 Improvements Using Clustering: Hierarchical Risk Parity

Here was used clustering from a practical problem in mean–variance style optimisation: if inputs are noisy, the optimiser can mistake noise for signal and over-allocate to whatever looks best in-sample. HRP avoids that fragility by side-stepping expected return estimation and by allocating risk across clusters rather than pushing directly through a covariance-matrix inverse.

$$d(i, j) = \sqrt{\frac{1 - \rho_{ij}}{2}}$$

This distance turns correlations into a clustering input. Assets that move similarly end up closer together, and hierarchical clustering can then group them before the recursive allocation step. The final weights are set by recursive bisection so that no single cluster dominates total risk.

Table 24: HRP in plain language

Stage	What happens
Distance step	Convert the correlation matrix into pairwise distances using $d(i, j) = \sqrt{(1 - \rho_{ij})/2}$.
Clustering step	Build a hierarchical tree so that similar assets sit in the same branch.
Quasi-diagonalisation	Reorder the covariance matrix so related assets sit next to each other.
Recursive bisection	Split the asset tree into sub-clusters and allocate weight inversely to cluster variance.
Practical effect	The portfolio spreads risk across groups instead of chasing the highest estimated mean return.

The resulting HRP weights are much more diversified than the constrained Kelly solution. The largest allocations are TLT (20.82%), GLD (15.66%), PG (10.22%), VNQ (9.40%), JNJ (9.39%), and XOM (8.71%). That is exactly the kind of pattern clustering is designed to produce: less concentration in a few equity names and more balance across different risk buckets.

6.7 Improvements Using Detoning

Even after denoising, the first eigenvector in the correlation matrix can still represent a broad market mode that pushes many assets in the same direction. If the goal is diversification, that common factor can make the optimiser look more confident than it should be about concentrated bets. Detoning removes that dominant mode and then re-normalises the matrix.

$$\mathbf{C}_{\text{detoned}} = \mathbf{C} - \lambda_{\text{mkt}} \mathbf{v}_{\text{mkt}} \mathbf{v}_{\text{mkt}}^\top$$

Table 25: What detoning changes

Question	Practical answer
What is removed?	The first eigenvector, interpreted as the broad market factor.
Why remove it?	It can dominate the covariance structure and hide genuine diversification opportunities.
What happened in our run?	The largest eigenvalue was 3.5170, equal to 29.3% of total trace before detoning.
What happens after denoising first?	The stacked denoised→detoned matrix still has a dominant market factor, with eigenvalue 3.5243 (29.4% of trace), which is then removed.
Expected portfolio effect	Slightly less concentration and a better chance that diversifying assets receive positive weight.

7 Step 5: Applying the ML Improvements

7.1 What was implemented

We applied the following improvements to the same 12-asset universe used in Step 1:

Table 26: Step 5 methodology menu

Tool	Core merit	Main drawback
HRP	Stable, cluster-based diversification without matrix inversion.	Does not explicitly maximise growth through expected returns.
Constrained Kelly	Targets long-run compounding growth while keeping a 20% cap and no shorting.	Still sensitive to estimation error in the inputs.
Double-Kelly	Pushes the growth logic further through leverage.	Can magnify drawdowns and tail losses very quickly.
Denoising	Cleans the covariance estimate by shrinking noise-like eigenvalues.	By itself, it may not change the final portfolio if the original cap constraints already bind.
Detoning	Removes the broad market mode and can open room for more diversification.	Can lower concentration but does not guarantee better raw return.

7.2 Explaining results

Table 27: Step 5 log excerpt reference

Logged item	Result from the script run
Denoising noise band	[0.6960, 1.3589]
Noisy eigenvalues	4 of 12
Market eigenvalue before detoning	3.5170 (29.3% of trace)
Market eigenvalue after denoising, before stacked detoning	3.5243 (29.4% of trace)

=== Denoising ===

MP noise band : [0.6960, 1.3589]

Noisy eigenvalues : 4 / 12

Denoised covariance matrix computed.

=== Detoning ===

Market eigenvalue : 3.5170 (29.3% of trace)

=== Denoised -> Detoned (stacked) ===

Market eigenvalue : 3.5243 (29.4% of trace)

7.3 Portfolio weights after the improvements

The denoising-only portfolio is practically identical to the baseline constrained Kelly solution: the same five assets remain pinned at the 20% cap. This is a useful result in itself, because it shows that the binding cap is already strong enough that a modest covariance cleaning step does not move the optimum. Detoning changes the solution slightly by allowing a small positive weight to AAPL and reducing MSFT below the cap. The stacked denoised+detoned version pushes this effect a bit further. HRP is the most visibly different portfolio because it spreads risk across all 12 assets.

Table 28: Step 5 portfolio weights (%)

Ticker	Kelly	Denoised	Detoned	Denoised+Detoned	HRP
AAPL	0.00	0.00	0.76	1.60	4.68
AMZN	20.00	20.00	20.00	20.00	2.94
BRK-B	20.00	20.00	20.00	20.00	6.04
GLD	20.00	20.00	20.00	20.00	15.66
JNJ	0.00	0.00	0.00	0.00	9.39
JPM	20.00	20.00	20.00	20.00	2.98
MSFT	20.00	20.00	19.25	18.40	5.47
NEE	0.00	0.00	0.00	0.00	3.68
PG	0.00	0.00	0.00	0.00	10.22
TLT	0.00	0.00	0.00	0.00	20.82
VNQ	0.00	0.00	0.00	0.00	9.40
XOM	0.00	0.00	0.00	0.00	8.71



Figure 8: Step 5 weights comparison across the baseline Kelly portfolio and the ML-enhanced alternatives.

7.4 Merits and sequence

We believe that these methods are better seen as complementary tools than as substitutes. That logic still holds after integration. A practical sequence is:

Table 29: Suggested sequence for the three improvements

Step	Reason
1	Denoise first, because it removes the most obvious covariance noise before any further transformation.
2	Detone next, because removing the market factor is cleaner once the raw noise has already been reduced.
3	Re-optimize Kelly on the cleaned matrix and benchmark the result against HRP and the original Kelly portfolio.
4	Use HRP as the diversification benchmark, especially if stability and drawdown control matter more than pure growth maximisation.

8 Step 6: Do Multiple Improvements Outperform?

8.1 Seven-strategy out-of-sample comparison

Here we compare the original three Kelly sizing rules with four ML-enhanced alternatives over the same out-of-sample window, October 1 to December 31, 2025. We compare cumulative return, annualised Sharpe ratio, Sortino ratio, Maximum Drawdown, and 95%-CVaR.

Table 30: Step 6 out-of-sample results: all seven strategies

Strategy	Cum. Return (%)	Sharpe	Sortino	Max Drawdown (%)	95%-CVaR (%)
Kelly	2.3877	0.3995	0.3948	−5.5364	−1.4394
Double-Kelly	4.8324	0.5872	0.5858	−10.7663	−2.8787
Half-Kelly	1.1868	0.0243	0.0239	−2.8076	−0.7197
Denoised-Kelly	2.3877	0.3995	0.3948	−5.5364	−1.4394
Detoned-Kelly	2.4907	0.4331	0.4287	−5.5098	−1.4341
Denoised+Detoned-Kelly	2.6064	0.4708	0.4668	−5.4797	−1.4281
HRP	2.8174	0.9033	0.8875	−2.5599	−0.7931

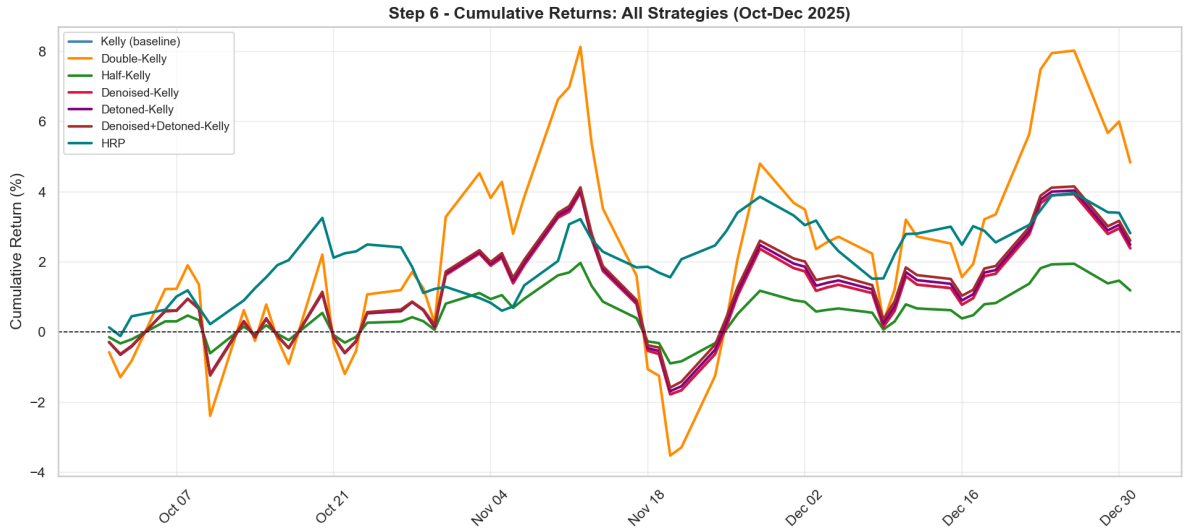


Figure 9: Step 6 cumulative returns for all seven strategies over the out-of-sample period.

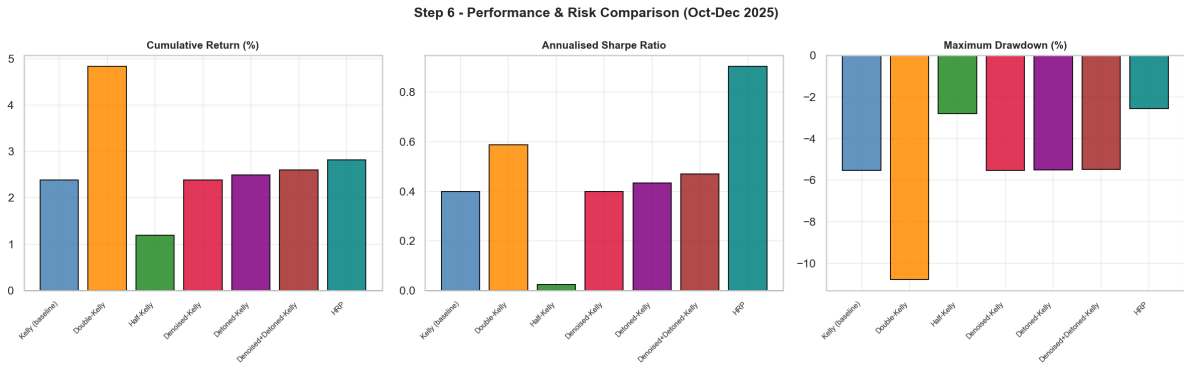


Figure 10: Step 6 comparison of cumulative return, annualised Sharpe ratio, and maximum drawdown across all seven strategies.

8.2 Interpretation of results

Table 31: Step 6 interpretation

Finding	Interpretation
Highest raw return	Double-Kelly remains the return leader at 4.83%, but that gain comes with the worst drawdown and the worst tail loss.
Best risk-adjusted result	HRP has the highest Sharpe ratio (0.9033), the highest Sortino ratio (0.8875), and the lowest maximum drawdown (−2.5599%).
Best Kelly-family improvement	Denoised+Detoned-Kelly is the strongest Kelly variant once ML adjustments are allowed, improving return and both risk-adjusted ratios relative to baseline Kelly.
Denoising alone	Denoised-Kelly is numerically the same as baseline Kelly in this sample because the weight cap still binds the same five assets.
Half-Kelly	Half-Kelly cuts risk materially, but the risk-free adjustment leaves its Sharpe ratio close to zero over this short test window.

As initial conclusion we think think that multiple improvements can help, but the value depends on what the investor is trying to improve. If the target is the highest raw return, Double-Kelly still wins. If the target is a better overall balance between return, drawdown, and tail risk, HRP is the clearest winner. If the investor wants to stay within the Kelly framework, denoising and detoning work better together than separately.

9 Conclusion

Table 32: Conclusion summary

Point	Summary
In-sample baseline result	With a 20% cap and no shorts, the original Kelly solution still concentrates in five assets at the cap.
Step 3 message	Walk-forward K-fold testing confirms that leverage lifts average return but also makes results much less stable across folds.
Step 4 message	Denoising, clustering, and detoning all try to reduce fragility, but they do so in different ways: cleaning noise, spreading risk across clusters, or removing the market mode.
Best ML-enhanced Kelly result	Denoised+Detoned-Kelly is the strongest Kelly-family variant in the Q4 2025 out-of-sample test.
Best overall trade-off	HRP gives the best overall risk-adjusted performance in the final seven-strategy comparison.

The traditional Kelly framework is useful, but its aggressive versions are highly sensitive to estimation error. Cross-validation shows that leverage can pay when the signal is right, yet the losses scale quickly when it is wrong. Among the ML-style improvements, denoising and detoning together improve the Kelly family modestly, while HRP changes the portfolio structure more substantially and delivers the best balance between return and risk in the final out-of-sample comparison.

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